

Skills Summary

Volumes by Known Cross Sections and Shells

Use this reference while completing your worksheet or when studying.

Known cross sections: $V = \int_a^b A(x) dx$ — slice the solid, write the AREA of one slice as a function of x , integrate. Slice areas: square $A = s^2$; semicircle on diameter d : $A = \frac{\pi}{8}d^2$; equilateral triangle of side s : $A = \frac{\sqrt{3}}{4}s^2$.

Cylindrical shells (about the y -axis): $V = \int_a^b 2\pi (\text{radius})(\text{height}) dx$, radius $= x$, height $=$ the vertical extent of the region at x .

Which method? Revolving about the y -axis while the region is described as $y = f(x)$ is the shell case — shells integrate in x , so f never has to be inverted. Washers in y would need x as a function of y first.

1. Solids with known cross sections

The base region only supplies the *length* of one side of each slice; the named shape turns that length into an area. Write s in terms of x first, then square it (every one of the three area formulas squares the side), then integrate over the x -range of the base.

Example: The base is the region under $y = x$ from $x = 0$ to $x = 3$. Cross sections perpendicular to the x -axis are squares. Find the volume.

Step 1. The slice area is what has to be integrated, and here the base gives the side of a square directly — the height of the region at x .

Step 2. $s = x$, so $A(x) = x^2$ and $V = \int_0^3 x^2 dx = \left[\frac{x^3}{3} \right]_0^3 = \frac{27}{3}$. The volume is **9** cubic units.

Try it: Same set-up with the base under $y = x$ from $x = 0$ to $x = 2$ and square cross sections. Find the volume.

check: $\frac{8}{3}$ cubic units

2. Cylindrical shells

Picture a thin vertical strip of the region at x . When the region turns about the y -axis the strip sweeps a hollow cylinder: circumference $2\pi x$, height $f(x)$, thickness dx . Multiply the three and integrate. The 2π is part of the model, not decoration — dropping it is the most common shell error.

Example: The region under $y = x$ from $x = 0$ to $x = 2$ is revolved about the y -axis. Find the volume by shells.

Step 1. The axis of revolution is the y -axis and the region is given as y in terms of x , so shells in x need no inversion.

Step 2. Radius x , height x , so $V = \int_0^2 2\pi x \cdot x dx = 2\pi \left[\frac{x^3}{3} \right]_0^2 = 2\pi \cdot \frac{8}{3}$. The volume is $\frac{16\pi}{3}$ cubic units.

Try it: The region under $y = x^2$ from $x = 0$ to $x = 2$ is revolved about the y -axis. Find the volume by shells.

check: 8π cubic units

3. Choosing and justifying the set-up

Ask three questions in order: what is the axis of revolution, how is the region described, and does a strip parallel to the axis reach across the whole region? Strips *parallel* to the axis give shells; strips *perpendicular* to it give discs or washers. If one choice forces you to invert the function, the other choice is usually the shorter route to the same number.

Example: The region under $y = x^2$ from $x = 0$ to $x = 1$ is revolved about the y -axis. Choose a method, justify it, and find the volume.

Step 1. The axis is the y -axis and the boundary is given as $y = x^2$; a vertical strip is parallel to that axis, so shells in x apply and no inversion to $x = \sqrt{y}$ is needed.

Step 2. $V = \int_0^1 2\pi x \cdot x^2 dx = 2\pi \left[\frac{x^4}{4} \right]_0^1 = \frac{2\pi}{4}$. The volume is $\frac{\pi}{2}$ cubic units.

Try it: The region under $y = x^3$ from $x = 0$ to $x = 1$ is revolved about the y -axis. Use shells.

check: $\frac{5}{2}\pi$ cubic units

(!) **Watch out:** for cross sections, integrate the *area* of a slice, never its side length — and check whether the base segment is the diameter or the radius of a semicircle, since mistaking one for the other changes every slice by a factor of four. For shells, keep the 2π and use the gap between the two boundary curves as the height when the region is bounded above and below.